Response to Reviewers

IEEE MINDS 2026 – Paper ID 958

Epileptic Seizure Detection Using Machine Learning: A Comparative Study of Class Imbalance Handling Techniques

Authors: Soha Chand, Atharv Dixit, Dr. Hima Deepthi Vankayalapati (MPSTME, SVKM's NMIMS)

We thank the reviewers for their careful reading and constructive comments. We addressed them in the revised camera-ready paper (6 pages). The main change is a new leave-one-patient-out (LOPO) evaluation over 22 CHB-MIT patients (809,187 windows) with statistical tests; the former two-patient study is kept as a preliminary study. Section and table numbers refer to the revised paper. Because the cross-dataset evaluation was redone, its numbers differ from the reviewed version. All values come from our runs.

Reviewer #1

Reviewer comment	Our response	Where changed
Interesting work; acceptable for presentation.	We thank the reviewer. No changes were requested.	–

Reviewer #2

Reviewer comment	Our response	Where changed
1. Only two training patients and one test patient; LOPO or k-fold over all 22 patients would strengthen the results.	Agreed. We added a LOPO evaluation over 22 CHB-MIT patients (chb01–20, 22, 23; chb21 excluded; 809,187 windows, 5,250 seizure windows). PCA and scaling are fitted on the 21 training patients only, and results are mean±std over the held-out patients with Wilcoxon and paired t-tests (Bonferroni-corrected). Main finding: SMOTE, Balanced RF and XGBoost do not improve AUC-ROC (0.802–0.810) or F1 (0.095–0.107) over a plain Random Forest (Wilcoxon $p \geq 0.77$ for AUC-ROC). The two-patient study is now labelled preliminary and its limits are stated.	Abstract; Sec. III-A, III-H; Sec. IV-A; Tables II, III; Fig. 2
2. Identical results for SMOTE+RF and SMOTETomek+RF should be explained or verified.	We verified this. With the sampling strategy used (minority raised to 5% of the majority), no Tomek links were found, so SMOTETomek removed no samples and produced the same training set as SMOTE (same seed, same results). This is now stated explicitly. SMOTETomek was therefore not rerun in the 22-patient evaluation.	Sec. III-E; Sec. IV-B
3. Very low recall under cross-dataset testing (e.g. 1.6% for Baseline RF) needs more explicit discussion given the cost of missed seizures.	We expanded the discussion. On the 8 Siena subjects used (464 seizure windows), the raw-feature models do not transfer (AUC-ROC 0.47–0.59; recall 0 for Baseline RF and SMOTE+RF). We discuss likely causes (adult vs paediatric population, equipment and montage, small number	Sec. IV-E; Table V; Sec. IV-G; Limitations

	of subjects) and say we did not isolate them. We also report event-level results: with the improved pipeline only 2 of 14 seizures are detected (5 of 14 with baseline-only normalisation, at 5.3 false alarms per hour). We state that the detector is not usable clinically.	
4. Extend the five-seed analysis to all techniques.	The five-seed analysis (seeds 42, 123, 456, 789, 999) now covers Baseline RF, SMOTE+RF, Balanced RF and XGBoost (mean±std; paired t-tests against the Baseline RF). SMOTETomek is omitted because it equals SMOTE (see 2). We also state that the seeds vary only model randomness on a fixed split; the main evidence is the 22-patient LOPO evaluation.	Sec. IV-B; Table IV
5. Benchmark against at least one deep-learning baseline.	We added two baselines on the same PCA features: a 1D-CNN and an MLP with positive-class weighting (two-patient setting; chb05 F1 0.148±0.014 and 0.239±0.064). Both are below the Random Forest. We note they are small and untuned and were not evaluated on 22 patients.	Sec. IV-F; Limitations

Reviewer #3 – Major issues

Reviewer comment	Our response	Where changed
Low F1 despite AUC-ROC of about 0.98; why is AUC high while F1 is low, and could another classifier or features give better F1?	We added an explanation and new experiments. AUC-ROC measures ranking over all thresholds; F1 is computed at one operating point, and with 0.65% seizure windows even a small false-positive rate produces many false alarms. A threshold study on chb05 shows F1 rising from 0.549 to 0.628 (threshold 0.50 → 0.40), and chb01 has AUC-ROC 0.963 yet no detection at 0.5. We also tested label-free improvements: per-patient normalisation, neighbouring-window context, temporal smoothing and alarm post-processing. These raised mean LOPO AUC-ROC from 0.802 to 0.891 (better in all 22 patients; $p < 0.001$) and F1 from 0.098 to 0.283, and 76 of 164 seizures (46%) were detected at 0.73 false alarms/h. With a seizure-free 30-minute baseline instead of whole recordings (more realistic) the values are AUC-ROC 0.868, F1 0.177, and 52 of 164 seizures (32%) at 0.70 false alarms/h. We state the caveats (the whole-recording variant is an upper bound; post-processing parameters have a mild leak).	Sec. III-F, III-G; Sec. IV-A, IV-C, IV-D; Tables II, III; Sec. IV-G
No LOPO validation; single fixed split (chb01, chb03 → chb05); single test patient.	See Reviewer #2, comment 1. LOPO over 22 patients is now the main evaluation. The F1 of 0.575 on chb05 is	Sec. IV-A; Sec. IV-G

	shown to be optimistic compared with the LOPO F1 of 0.10–0.28 for the same techniques.	
Limited cross-dataset recall; no threshold adjustment or domain adaptation.	We added (i) a threshold sweep on Siena (an oracle upper bound, since it uses the Siena labels): the best Baseline RF F1 is only 0.082 at threshold 0.10 (precision 0.058, recall 0.136); (ii) per-subject normalisation of the Siena features, a simple domain-adaptation step, which raised AUC-ROC from 0.585 to 0.749, and with context and smoothing to 0.771; and (iii) event-level results (2 of 14, or 5 of 14, seizures). Recall and F1 remain low and we say so. Full domain adaptation is left as future work.	Sec. IV-E; Table V
No comparison with deep-learning baselines.	See Reviewer #2, comment 5 (1D-CNN and MLP added).	Sec. IV-F
PCA: is it per-window or global, and is there leakage?	Clarified. PCA (with a StandardScaler) is fitted once on the training windows only, with 120 components, and then applied to the held-out windows. In LOPO the scaler and PCA are refitted on the 21 training patients for each fold. For Siena, the Siena features are scaled with their own scaler or per-subject normalisation, and the models are not retrained. The PCA ablation (40, 60, 90, 120 components) is reported.	Sec. III-C; Sec. IV-D
No statistical significance testing between techniques.	Added. Wilcoxon signed-rank and paired t-tests across the 22 patients against the Baseline RF, with Bonferroni correction (24 Wilcoxon tests, $\alpha = 0.0021$). The significant results are the FDR/hour differences of Balanced RF and XGBoost and the AUC-ROC gains of smoothing and the normalised variants. The five-seed study uses paired t-tests over seeds.	Sec. III-H; Table III; Sec. IV-A
No code or data availability statement.	Added. CHB-MIT and Siena are public on PhysioNet [4], [14]; the analysis code can be obtained from the authors on request.	End of Sec. V

Reviewer #3 – Minor issues

Reviewer comment	Our response	Where changed
Section headings use Arabic numerals.	Main section headings use Roman numerals (I–V), in IEEE style.	Whole paper
Captions should be above tables/figures.	Table captions are above the tables, as in the IEEE template. Figure captions are below the figures, as IEEE format requires.	Tables, figures
Abstract should be italicised.	The abstract is italic.	Abstract
Table I missing “Description” column header alignment.	Corrected.	Table I
Table V check mark (√) may not render.	Removed; Table V now contains numbers only (best values in bold).	Table V
Reference [11] (imbalanced-learn) is	The text now states that SMOTE,	Sec. III-E

cited but not discussed.	SMOTETomek and Balanced RF come from imbalanced-learn [11].	
Language issues (“the ratio comes out to be the ratio of 115.7:1”).	The sentence was rewritten (e.g. “The imbalance ratio is 115.7:1 in the two-patient study and 40:1 in the leave-one-patient-out evaluation”), and the text was proofread.	Sec. III-E
Flask/Chart.js front-end is not relevant.	Not part of this paper and not mentioned; no change needed.	–

Summary of limitations we state in the paper: only a subset of each CHB-MIT recording was used, so false-alarm rates are not rates for continuous monitoring; the whole-recording normalisation is an optimistic upper bound; post-processing parameters were chosen on other patients’ out-of-fold predictions, which gives a mild leak; imbalance techniques were not rerun on the normalised inputs; the Siena test uses 8 subjects only; and detection performance remains insufficient for clinical use.